Parikshith Y S

Robotics & AI Engineer | ROS2, PX4, Embedded AI, Edge Inference

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PROFILE

AI Engineering student at **Technische Hochschule Ingolstadt** with over 5 years of professional experience as an Embedded Engineer. Specializing in end-to-end **ROS2-based robotic systems** and **Edge AI solutions** including running Vision Language Models on resource-constrained hardware. Expertise bridges complex firmware (C/C++, FreeRTOS) and high-level autonomous behaviors (Python, PyTorch). Having supported **NVIDIA's DRIVE OS** and built autonomous drone solutions at **Hilti**, driven to create production-grade intelligent systems for real-world mobility and logistics.

WORK EXPERIENCE

Robotics & AI Engineer (Work Student)

Dec 2024 - Present

Hilti | Kaufering, Germany

- Architecting an end-to-end **ROS2**-based robotic system for autonomous construction logistics and tool tracking.
- Developed robust localization pipeline using **Particle Filters** for real-time tool tracking in complex environments.
- Integrating **Gaussian Processes** for intelligent path planning and environment modeling to enhance drone navigation efficiency.
- Designing and testing **PX4**- and **ROS2**-based drone control systems with autonomous waypoint navigation and return-to-launch features.
- Integrating computer vision (**LiDAR** + onboard cameras) for obstacle detection and environment awareness.
- Developing and debugging flight behaviors in **Gazebo**/**SITL** before hardware deployment.
- Supporting research on **Edge AI** applications - lightweight **deep learning** models for real-time decision-making on embedded systems.

Senior Embedded Engineer (Contractor)

Jul 2023 - Jul 2024

NVIDIA via Vayavya Labs | Bengaluru, India

- Supported **DRIVE OS** component of NVIDIA's autonomous driving software stack.
- Took ownership of software development for an entire module, spanning **pre-silicon validation**, **FPGA bring-up**, and **post-silicon integration** with end-to-end visibility across the development pipeline.
- Collaborated with internal software groups to design and implement the base software layer (**Linux**/**QNX** OS, device drivers).
- Developed hardware-specific drivers using **AutoSAR** firmware for NVIDIA's upcoming **ADAS** hardware.
- Conducted driver validation through functional test cases; resolved customer-reported **DRIVE OS** bugs.
- Matured prototype software to production-quality standards.

Embedded Systems Software Developer (Contract)

Apr 2023 - Jul 2023

Stasis Labs | Bengaluru, India

- Decoded **medical device** data using manufacturer communication protocol specifications.
- Designed and developed application software to read, format, and transmit data from **medical devices**.
- Collaborated with hardware team on cable design and **signal-level validation** for reliable data transmission.
- Wrote automation scripts for task scheduling, device simulators, and test rigs.

Embedded Engineer

Aug 2021 - Oct 2022

Frenustech | Bengaluru, India

- Developed FPGA BSP for a custom **RISC-V** core; implemented and validated **I2C, UART, SPI** drivers on Xilinx ARTY A7.
- Implemented real-time **object detection** on STM32H747I (dual-core ARM) using a compressed **TensorFlow Lite** model.
- Developed **FreeRTOS**-based applications and **multi-sensor fusion** firmware for STM32 microcontrollers.
- Designed SDK for ARM-based microcontrollers; conducted performance benchmarking.

UAV Researcher Intern

Mar 2021 - Sep 2021

Vyomastra | Bangalore, India

- Developed **UAV firmware** for photogrammetry and AI-based drone applications.
- Designed UAV subsystems: **battery management**, **flight controller**, and **NPNT** compliance modules.

TECHNICAL SKILLS

Robotics & Autonomy
ROS2, PX4, Gazebo, NVIDIA Isaac Sim, NVIDIA Omniverse, Isaac ROS, OpenCV, SLAM, Particle Filters, Path Planning

AI & Edge Inference
PyTorch, TensorFlow Lite, ONNX, TensorRT, INT8 / FP16 Quantization, Model Pruning, DeepStream, Edge AI, VLMs

Embedded & Systems
C / C++, Embedded C, AutoSAR, FreeRTOS, Python, Linux, QNX, STM32, Git, Jupyter

EDUCATION

M.Eng. - AI Engineering of Autonomous Systems
Oct 2024 - Present

Technische Hochschule Ingolstadt | Germany
Thesis: Enhancing UAV Control with Large Language Models: Natural Language Interfaces and Context-Aware Guidance

B.Eng. - Electronics and Communication Engineering
Aug 2017 - Aug 2021

Vidyavardhaka College of Engineering | Mysuru, India
Final Grade: 1.7

PROJECTS

Autonomous Object Sorting & Transport - Dual-Robot Collaboration
Apr 2025 - Jul 2025

- DJI RoboMaster EP + Arduino Braccio arm for warehouse-style object sorting and transportation.
- ArUco marker + OpenCV navigation - **85% alignment accuracy**.
- HSV color detection: **95% (red)** and **90% (green)** under varying lighting.
- Multi-threaded control across Raspberry Pi and Windows PC with real-time sync.

Custom UAV Flight Controller & Transceiver Stack Final Year Project
Oct 2020 - Apr 2021

- Designed full UAV flight controller firmware from ground up: sensor fusion, control loops, and RF comms.
- IMU sensor fusion (accel, gyro, magnetometer) for real-time attitude estimation.
- PID attitude stabilization and altitude hold tuned through iterative flight tests.
- Custom RF transceiver for real-time telemetry and remote command reception.

GSM-Based Real-Time Location Tracking System
Jan 2019 - Mar 2019

- GPS + GSM real-time asset tracking with cellular SMS-based location transmission.
- Microcontroller firmware to parse NMEA sentences and format data for reliable output.

LANGUAGE SKILLS

English — C1, Proficient

HONOURS & AWARDS

National Level Hackathon Winner | ATME Engineering College — Mar 2019
National Level Hackathon Winner | PES College, Mandya — Oct 2019